1. Clinical Question & Core Hypothesis

Clinical Question

NMIBC frequently recurs but only a subset progress to muscle-invasive disease; why do some tumors with abundant stroma/immune cells still progress? The Ontologist highlights TGF-β–driven stromal remodeling, immune contexture (Immunoscore/CD8), and CTC/ liquid readouts as core axes, suggesting interplay between fibrosis and immune exclusion as a determinant of progression. 2) Tier 1 paths used: I built on the TGF-β (TGFB1) path because it mechanistically links to stromal remodeling and immune suppression, and on the Immunoscore axis (operationalized here via CD8A) because CD8+ T cell presence/function mediates anti-tumor control; these two provide a direct mechanistic contrast (fibrotic exclusion vs cytotoxic immunity). 3) Tier 2 evidence that supports the claim: urine DNA methylation and TMB add complementary noninvasive/genomic context for tumor aggressiveness; E-FABP and metabolic markers (HDL-C/CDKN2A) provide orthogonal metabolic/metastatic modifiers; CTCs can validate dissemination potential. I treat these as secondary modifiers that could modulate the core TGFB1–CD8A–TSR interaction. 4) Endpoint choice rationale: Progression-free interval (PFI) is the most problem-relevant clinical endpoint for NMIBC (key clinical question is which non–muscle-invasive tumors will progress), so PFI offers mechanistic and translational clarity. 5) Deliberate exclusions: I excluded AI-accessibility and SARIFA as core mechanistic variables (methodological/exploratory), and kept urine methylation, TMB, E-FABP, HDL-C/CDKN2A, and CTCs in secondary_modifiers to avoid diluting the causal TGFB1–stroma–CD8A hypothesis. I also excluded Tier 3 exploratory signals from the core hypothesis to maximize focus and testability?

Working Hypothesis

Among non–muscle-invasive bladder cancer (NMIBC) patients, tumors with a high stromal-to-tumor ratio (TSR) that also show elevated TGFB1 signaling will exhibit a fibrotic ECM and an immunosuppressive microenvironment characterized by reduced CD8A+ T cell infiltration or activity within tumor compartments. This TGFB1–CAF/ECM axis mediates immune exclusion and promotes tumor progression, resulting in a shorter progression-free interval (PFI). The association is strongest in TSR-high tumors with high TGFB1 and low CD8A expression; in these tumors, secondary modifiers (urine DNA methylation, tumor mutational burden, circulating tumor cells [CTCs], E-FABP, CDKN2A status, HDL-C) reflect overall aggressiveness and metastatic potential but do not alone drive the core mechanistic pathway. The model predicts: (1) TSR-high/TGFB1-high/CD8A-low tumors have the worst PFI; (2) spatial profiling will reveal CD8+ T cell exclusion from tumor nests and intact/aged stroma; (3) disrupting TGFB1 signaling or CAF–ECM remodeling will enhance CD8A+ T cell localization and function and improve PFI; (4) urinary and CTC-based modifiers will correlate with risk but not abolish the TGFB1–stroma–CD8A axis.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	TGFB1, CD8A
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TGFB1: TGFB1) path because it mechanistically links to stromal remodeling and immune suppression, and on the Immunoscore axis (operationalized here via CD8A) because CD8+ T cell presence/function mediates anti-tumor control; these two provide a direct mechanistic contrast (fibrotic exclusion vs cytotoxic immunity).

CD8A: CD8A) because CD8+ T cell presence/function mediates anti-tumor control; these two provide a direct mechanistic contrast (fibrotic exclusion vs cytotoxic immunity).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CTCs	Supporting modifier from knowledge graph
Secondary	urine_DNA_methylation	Supporting modifier from knowledge graph
Secondary	TMB	Supporting modifier from knowledge graph
Secondary	E-FABP	Supporting modifier from knowledge graph
Secondary	CDKN2A	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

Prediction 2: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CTCs: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- urine_DNA_methylation: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.